GRID-Q

A Hybrid AI–Quantum Framework for Resilient Distribution Grid Expansion Planning

A Closed-Loop Grid Optimization System for Congestion-Aware Infrastructure Planning using AI, QUBO Optimization, Quantum Sampling, and Fujitsu Quantum Simulator

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Final Report Submission

Participant Name: J K Pawan Kumar and Tarun Solanki
User ID: g155-user1 and g155-user2
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Executive Summary

Problem

Distribution-grid expansion planning is a coupled combinatorial problem involving congestion, voltage stress, outage exposure, renewable variability, demand growth, and budget limits. With 50 binary upgrade candidates, each network contains $2^{50} = 1.125 \times 10^{15}$ possible infrastructure plans, making exhaustive evaluation impractical.

Innovation

GRID-Q introduces a hybrid AI–quantum planning framework integrating:

- power-flow simulation and hard stress-scenario generation,
- physics-grounded QUBO formulation,
- AI-guided search-space compression,
- quantum feature-map scoring and calibrated QAOA-style sampling,
- safety-gated Quantum Large-Neighborhood Search,
- distributed mpiQulacs simulation using Fujitsu QARP.

Key Results

GRID-Q executive results summary.

Metric	Result
Raw planning space	$2^{50} = 1.125 \times 10^{15}$ plans
AI variable reduction	90–92%
Maximum search compression	70.37 trillion-fold
AI difficulty predictor	0.98 accuracy, 0.989 F1, 1.0 ROC-AUC
14-qubit feasible-plan probability	0.1967–0.8000
Q-LNS robustness runs	450 fresh runs
Safety acceptance / accepted loss	96.67% / 0.00%
Significant energy/oracle-regret cells	6 benchmark cells
Largest mpiQulacs run	30 qubits on 16 MPI ranks
Resource boundary	32 qubits on 32 MPI ranks
Congestion / outage / resilience gains	44.26% / 35.50% / 42.31%
Stable-topology validation pass rate	100% over 3 stable topologies
IEEE69 status	Collapse-boundary diagnostic

No single method solves the full planning challenge efficiently. Power-flow simulation provides physical realism, AI compresses the intractable search space, QUBO formulation makes the problem quantum-compatible, calibrated QAOA-style sampling explores feasible plans, and safety-gated Q-LNS refines high-impact local decisions without degrading physical validation.

The 10–14 qubit quantum neighborhoods are targeted refinement subproblems inside the original 50-variable planning space, not replacements for the full problem.

Fujitsu QARP

Fujitsu mpiQulacs/QARP enabled distributed `multi-cpu` statevector execution, multi-rank simulation, scaling to 30 qubits on 16 MPI ranks, and identification of a 32-qubit resource boundary. FusedSWAP was successfully invoked through `qulacs.circuit.QuantumCircuitOptimizer`; the observed overhead provides useful feedback on circuit-layout sensitivity.

Business Impact

GRID-Q improves congestion, outage-risk, and resilience proxy metrics while supporting CAPEX-aware planning and cross-topology validation. These are planning proxies, not audited utility financial values, but they connect quantum-enabled optimization to enterprise infrastructure decision-making.

Final Takeaway

GRID-Q demonstrates a practical quantum-HPC decision workflow for resilient grid expansion planning, not merely a simulated QAOA benchmark.

1. Abstract

Grid expansion planning is a high-stakes combinatorial optimization problem involving congestion, voltage stability, reliability, budget limits, and infrastructure investment decisions. We present **GRID-Q**, a hybrid AI–quantum framework for scalable distribution-grid expansion planning under stressed operating conditions. GRID-Q integrates power-flow simulation, hard-instance generation, QUBO formulation, AI-guided candidate filtering, calibrated QAOA-style quantum sampling, safety-gated Quantum Large-Neighborhood Search, and Fujitsu mpiQulacs/QARP execution and scaling analysis into one end-to-end optimization pipeline.

GRID-Q was evaluated on IEEE33, IEEE69, IEEE118, and synthetic California grid networks under demand spikes, renewable variability, line derating, and compound stress. The system generated 50-variable grid-expansion QUBOs, corresponding to $2^{50} = 1.125 \times 10^{15}$ possible infrastructure plans per network. AI filtering compressed these raw search spaces to 4–5 variable quantum-ready subproblems, achieving **90–92% variable reduction** and up to **70.37 trillion-fold search-space compression**. The AI difficulty predictor achieved **0.98 accuracy**, **0.989 F1**, and **1.0 ROC-AUC**; these scores are used for routing and search-space reduction, not as standalone proof of planning optimality.

For the quantum layer, GRID-Q uses quantum feature-map scoring, calibrated QAOA-style sampling, and physics-grounded Quantum Large-Neighborhood Search. Candidate benefits and pairwise synergies are evaluated using power-flow response, while AI uncertainty routes high-impact decisions into quantum neighborhoods. In robustness testing, GRID-Q ran **450 fresh Q-LNS trials** across stable grid benchmarks, achieving **96.67% safety acceptance**, **0.00% accepted loss rate**, positive mean QUBO-energy improvement, positive mean physical grid-gain improvement, and statistically significant energy/oracle-regret improvements in **6 benchmark cells**. At 14 qubits, calibrated quantum sampling maintained feasible-plan probability between **0.1967** and **0.8000** over **16,384-plan** search spaces.

GRID-Q was evaluated for scaling on **Fujitsu mpiQulacs/QARP**, including distributed state-vector execution, multi-rank simulation, and communication-aware circuit optimization using FusedSWAP. GRID-Q verified `multi-cpu` distributed execution and successfully scaled to **30 qubits on 16 MPI ranks**, while a **32-qubit run on 32 MPI ranks** established the current resource boundary. FusedSWAP optimization was successfully invoked through `qulacs.circuit.QuantumCircuitOptimizer`; on the tested stress circuits it introduced overhead rather than acceleration, providing useful communication-aware QARP feedback.

The result is not merely a QAOA demonstration, but a full hybrid AI–quantum infrastructure optimizer combining grid physics, AI uncertainty, QUBO modeling, calibrated quantum sampling, Fujitsu quantum simulation infrastructure, and safety-gated quantum refinement for realistic grid expansion planning.

2. Background and Objectives

Modern distribution grids are becoming harder to plan because of rapid electrification, renewable-energy variability, aging infrastructure, demand uncertainty, and increasingly severe operating stresses

[1, 2]. Utilities must decide where to reinforce lines, add tie-lines, upgrade transformers, or modify switching configurations while satisfying congestion, reliability, voltage, resilience, and budget constraints. These decisions are strongly coupled: a reinforcement that improves one part of the network can shift congestion, voltage stress, or outage exposure elsewhere.

Distribution-grid expansion is therefore a naturally combinatorial planning problem. Each candidate infrastructure action can be represented as a binary decision, producing a search space that grows exponentially as 2^n [5, 10]. As the number of candidate actions increases, exhaustive evaluation becomes impractical, and classical heuristics may struggle to preserve solution quality across stressed operating scenarios, uncertain demand, renewable fluctuations, and outage conditions [4].

This motivates optimization frameworks that can combine power-system realism with scalable search-space reduction and advanced combinatorial optimization. Quantum-compatible formulations such as QUBO provide a natural representation for binary infrastructure decisions, while AI-guided filtering can reduce the effective decision space before quantum or hybrid refinement is applied [6, 12]. In this context, hybrid AI–quantum optimization offers a promising direction for identifying high-value grid-expansion plans under complex constraints.

The objective of **GRID-Q** is to develop and evaluate a hybrid AI–quantum framework for resilient distribution-grid expansion planning. The work focuses on converting realistic grid-planning scenarios into QUBO optimization problems, reducing the candidate decision space using AI-guided filtering, applying quantum-compatible sampling and refinement to selected high-impact neighborhoods, and validating final plans using power-flow and business-facing metrics.

The main objectives of this work are:

1. Build realistic distribution-grid congestion and expansion-planning instances using power-flow simulation, multiple grid topologies, and stressed operating scenarios.

2. Formulate grid expansion as a QUBO problem that captures congestion relief, build cost, outage risk, voltage penalties, budget limits, reliability terms, and coupled infrastructure interactions.

3. Develop an AI-guided filtering layer that reduces the effective combinatorial search space while preserving high-value candidate reinforcements.

4. Apply calibrated quantum-compatible sampling and quantum local-neighborhood refinement to targeted subproblems within the larger grid-expansion search space.

5. Evaluate solution quality using congestion reduction, outage-risk reduction, resilience improvement, CAPEX efficiency, feasibility, and post-power-flow validation.

6. Compare the proposed hybrid approach against classical baselines using ablations, robustness tests, cross-topology generalization, and statistical significance analysis.

7. Validate Fujitsu/QARP utilization through mpiQulacs-based distributed simulation, multi-rank execution, qubit-scaling experiments, and communication-aware circuit evaluation.

8. Identify the complexity regimes where hybrid AI–quantum optimization provides the strongest measurable value over classical-only planning methods.

3. System Overview

GRID-Q is an end-to-end hybrid AI–quantum framework for resilient distribution-grid expansion planning. The system converts stressed grid-planning scenarios into optimization-ready decision problems, represents candidate infrastructure actions as binary variables, reduces the effective search space using AI-guided filtering, formulates the selected decision space as a QUBO, applies quantum-compatible sampling and local refinement, and validates proposed expansion plans using power-flow and business-facing metrics.

The overall workflow is shown conceptually in Figure 1.

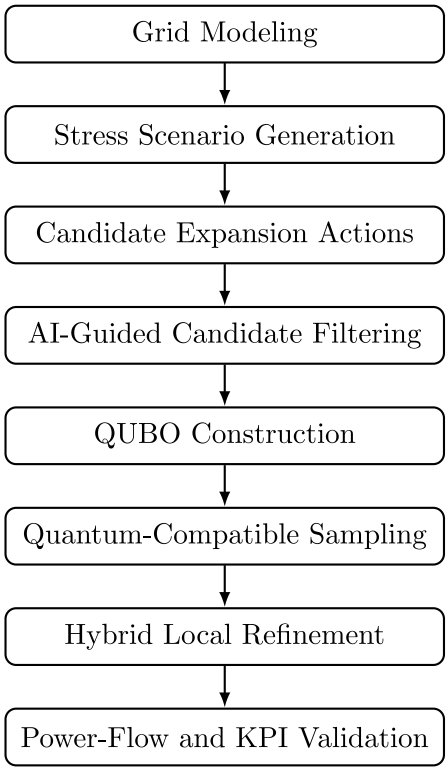

Figure 1: High-level GRID-Q system workflow from grid modeling to AI-guided QUBO construction, quantum-compatible sampling, hybrid refinement, and validation.

3.1. Architecture

GRID-Q is organized into six functional layers:

1. **Grid Modeling Layer:** Builds distribution-grid planning instances and evaluates operating conditions such as line loading, voltage behavior, congestion severity, and unmet demand.

2. **Stress Scenario Layer:** Generates challenging operating scenarios including demand growth, renewable variability, line outages, correlated overloads, and compound congestion events.

3. **Candidate Expansion Layer:** Defines possible infrastructure actions such as line reinforcements, tie-lines, feeder upgrades, transformer upgrades, and switching-related actions. Each candidate action is represented as a binary decision variable.

4. **AI-Guided Filtering Layer:** Scores and filters candidate expansion actions using grid-derived features, predicted value, uncertainty, and scenario sensitivity. This layer reduces the effective decision space before quantum-compatible optimization is applied.

5. **Quantum Optimization Layer:** Converts the filtered planning problem into a QUBO representation and applies quantum-compatible sampling and local-neighborhood refinement to explore candidate expansion plans.

6. **Validation Layer:** Evaluates proposed plans using post-optimization checks, power-flow validation, feasibility constraints, and business-facing planning metrics.

3.2. Decision Representation

Each candidate expansion action is encoded as a binary variable:

$$x_i \in \{0, 1\},$$

where $x_i = 1$ means that candidate action i is selected, and $x_i = 0$ means that the action is not selected.

The resulting expansion-planning problem is represented as a QUBO:

$$\min_x x^T Q x,$$

where Q captures the value, cost, penalties, and interaction structure of candidate grid upgrades.

The QUBO objective combines engineering and planning terms such as:

$$C_{\text{congestion}} + C_{\text{build}} + C_{\text{outage}} + C_{\text{voltage}} + C_{\text{switching}} + P_{\text{budget}} + P_{\text{reliability}} + P_{\text{connectivity}}.$$

This representation allows GRID-Q to model grid expansion as a quantum-compatible binary optimization problem while preserving engineering-relevant constraints.

3.3. Hybrid AI–Quantum Workflow

The system is hybrid by design. The AI layer first identifies promising and uncertain candidate actions from the full planning space. The filtered candidate set is then converted into a QUBO and passed to the quantum-compatible optimization layer. Quantum sampling explores candidate bitstrings from the QUBO landscape, while local refinement focuses on high-impact neighborhoods within the original grid-expansion problem.

The final decision is not selected from QUBO energy alone. Candidate plans are checked through validation layers that consider feasibility, grid behavior, cost, congestion relief, reliability, and resilience. This makes GRID-Q a planning-oriented decision framework rather than only a standalone quantum optimization experiment.

3.4. Fujitsu/QARP Integration

GRID-Q uses Fujitsu mpiQulacs/QARP as the quantum simulation backend for distributed statevector execution and quantum-circuit evaluation. The Fujitsu/QARP layer supports simulation of quantum-compatible optimization circuits, multi-rank execution, and analysis of communication-aware circuit behavior.

In the overall system, Fujitsu/QARP is used to evaluate whether the quantum optimization layer can scale beyond small local tests and remain connected to realistic grid-planning subproblems. Detailed scaling numbers, resource-boundary findings, and optimizer-behavior results are reported later in the Results and Fujitsu/QARP Utilization sections.

3.5. Validation Philosophy

GRID-Q validates expansion plans using both technical and planning-level criteria. The system evaluates whether selected upgrades improve grid behavior, remain feasible under stressed conditions, and produce measurable value under business-facing metrics. The validation layer therefore links optimization outputs to practical infrastructure-planning goals such as congestion reduction, outage-risk reduction, resilience improvement, and cost-aware decision quality.

Detailed numerical results, ablation studies, robustness tests, topology-generalization outcomes, statistical significance analysis, and claims-to-evidence mapping are reported in the Results section.

4. Methodology

The methodology of **GRID-Q** transforms distribution-grid expansion planning into a hybrid AI–quantum optimization workflow. The system combines power-flow simulation, stress-scenario generation, candidate expansion modeling, physics-grounded QUBO formulation, AI-guided search-space reduction, calibrated quantum-compatible sampling, Quantum Large-Neighborhood Search, hybrid ensemble selection, Fujitsu mpiQulacs/QARP execution, and post-power-flow validation.

The goal of the methodology is to ensure that candidate expansion plans are not evaluated only as abstract binary solutions, but also through engineering and planning criteria such as congestion relief, resilience, outage-risk reduction, feasibility, and cost-aware infrastructure value.

4.1. Grid Simulation and Congestion Modeling

The first stage constructs grid-planning environments using power-flow simulation. Multiple network topologies are evaluated to represent different operating structures and stress behaviors. For each network, GRID-Q computes engineering-relevant indicators such as line loading, voltage violations, congestion severity, thermal overloads, and unmet demand.

This stage prevents the optimization problem from becoming a toy binary benchmark. Instead, candidate decisions are derived from physically meaningful grid conditions and stressed operating regions where infrastructure upgrades may be required.

4.2. Hard Instance Generation

GRID-Q generates hard planning instances by applying stress regimes such as demand spikes, renewable fluctuations, line failures, correlated overloads, and compound congestion events. These stress cases increase coupling between grid regions and create scenarios where local infrastructure decisions can have network-wide effects.

This step is important because distribution-grid expansion is not only a static cost-minimization problem. A plan that performs well under normal conditions may fail under uncertainty, outages, or renewable intermittency. Therefore, GRID-Q evaluates candidate plans under progressively harder regimes.

4.3. Candidate Expansion Graph

Each grid instance is converted into a candidate expansion graph. Candidate actions include line reinforcements, tie-lines, feeder upgrades, transformer upgrades, and other topology-improving infrastructure decisions. Each candidate action is represented as a binary variable:

$$x_i \in \{0, 1\},$$

where $x_i = 1$ means that candidate action i is selected, and $x_i = 0$ means that it is not selected. For a candidate set of size n, the raw search space is:

$$2^n.$$

This exponential scaling motivates AI-guided filtering and quantum-compatible optimization rather than exhaustive enumeration.

4.4. Physics-Grounded QUBO Formulation

The grid expansion problem is formulated as a Quadratic Unconstrained Binary Optimization problem:

$$\min_x x^T Q x.$$

The QUBO matrix Q encodes both individual candidate value and pairwise interaction structure between infrastructure decisions. GRID-Q combines the following planning terms:

$$C_{\text{total}} = C_{\text{congestion}} + C_{\text{build}} + C_{\text{outage}} + C_{\text{voltage}} + C_{\text{switching}} + P_{\text{budget}} + P_{\text{reliability}} + P_{\text{connectivity}}.$$

Here, $C_{\text{congestion}}$ rewards actions that reduce overloads, C_{build} penalizes expensive reinforcements, C_{outage} captures reliability exposure, C_{voltage} penalizes voltage instability, and the penalty terms encode planning constraints such as budget, reliability, and connectivity.

In the physics-grounded version of the QUBO, candidate benefits and pairwise synergies are evaluated using power-flow response. For candidate pairs, synergy is measured as:

$$\text{synergy}_{ij} = \text{benefit}_{ij} - \text{benefit}_i - \text{benefit}_j.$$

This allows GRID-Q to distinguish complementary upgrades from redundant or conflicting infrastructure actions.

4.5. AI-Guided Search-Space Reduction

Before quantum-compatible optimization, GRID-Q applies an AI-guided filtering layer. This layer scores candidate infrastructure actions using features such as congestion severity, benefit proxy, estimated cost, graph connectivity, stress score, coupling score, and reliability relevance.

The role of AI is not to replace optimization. Instead, the AI layer acts as a routing and compression mechanism that identifies high-value or uncertain regions of the planning space. These selected candidates are then passed to the QUBO and quantum-compatible optimization stages.

To reduce data leakage, AI validation is performed using splits by network and stress scenario rather than random row-level splits. Candidate records from the same grid scenario are not allowed to appear in both training and test sets. This ensures that the AI layer is evaluated on unseen planning conditions rather than memorizing candidate-level patterns from the same scenario.

Because the benchmark features are structured, the AI layer is treated as a search-space reduction tool, not as final proof of planning optimality. Final acceptance still depends on QUBO evaluation, quantum/local refinement, ensemble selection, and post-power-flow validation.

4.6. Quantum Optimization using Fujitsu QARP and mpiQulacs

After AI filtering and QUBO construction, GRID-Q applies calibrated QAOA-style quantum-compatible sampling over the QUBO energy landscape. Candidate bitstrings are evaluated using objective quality, feasibility, elite-solution concentration, and distributional structure.

The quantum layer is designed to support hybrid refinement rather than standalone quantum replacement. Quantum sampling explores targeted, uncertain, high-impact subproblems selected by the AI layer, while physical feasibility is preserved through post-power-flow validation.

The implementation is aligned with Fujitsu QARP capabilities, including distributed statevector execution through mpiQulacs, multi-rank simulation, and communication-aware circuit evaluation. Distributed execution is checked using:

```
QuantumState(n_qubits, use_multi_cpu=True)
```

and:

```
state.get_device_name().
```

Detailed scaling outcomes, qubit limits, runtime behavior, and communication-aware optimizer observations are reported separately in the Results and Fujitsu/QARP Utilization sections.

The quantum sampling process is summarized in Figure 2.

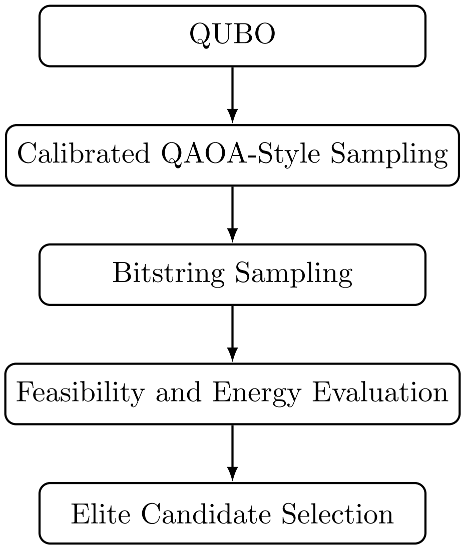

Figure 2: Quantum-compatible sampling workflow used in GRID-Q.

4.7. Quantum Large-Neighborhood Search

To connect quantum-compatible sampling with larger planning problems, GRID-Q implements Quantum Large-Neighborhood Search. Instead of attempting to solve the full planning problem directly in one quantum circuit, Q-LNS selects high-impact local neighborhoods using AI uncertainty, stress, and coupling features.

The Q-LNS process is:

1. build an initial classical feasible expansion plan,
2. identify uncertain high-impact local neighborhoods,
3. freeze the remaining decision variables,
4. solve the selected local subproblem using calibrated quantum-compatible sampling,
5. decode elite sampled bitstrings into candidate upgrade plans,
6. re-run post-power-flow validation on each proposal,
7. accept a proposal only if it is non-worse than the baseline under safety checks,
8. repeat across refinement rounds and random seeds.

This design allows the quantum layer to act as a controlled local improvement operator while preserving the full planning context and physical validation requirements.

4.8. Hybrid Ensemble Selection

GRID-Q combines AI-generated, classical, and quantum-sampled solutions using a hybrid ensemble layer. The ensemble selects robust expansion plans by considering objective value, feasibility, resilience, stress behavior, and planning constraints.

Algorithm 1 Safety-Gated Quantum Large-Neighborhood Search

Require: Grid QUBO, candidate grid actions, power-flow validator
Build an initial feasible classical expansion plan.
Score candidates using AI value, uncertainty, stress, and coupling features.
Select a high-impact quantum neighborhood.
Freeze all variables outside the selected neighborhood.
Run calibrated QAOA-style sampling on the local QUBO.
Decode elite sampled bitstrings into candidate upgrade plans.
Re-run post-power-flow validation on each proposal.
Accept a proposal only if physical grid performance is non-worse than baseline.
Repeat across refinement rounds and random seeds.
return safety-gated hybrid expansion plan.

This stage is included because no single solver is expected to dominate every regime. Classical search can perform well on smaller or easier instances, AI improves tractability by filtering the search space, and quantum-compatible sampling explores uncertain local neighborhoods. The ensemble layer reduces dependence on any single method and improves reliability under stressed scenarios.

Ablation results that quantify the contribution of each component are reported separately in the Results section.

4.9. Safety-Gated Post-Power-Flow Acceptance

Quantum-generated proposals are not accepted directly from QUBO energy alone. GRID-Q uses a safety gate that accepts a Q-LNS proposal only when post-power-flow evaluation is non-worse than the baseline.

The safety gate checks:

- budget feasibility,
- selected-count feasibility,
- non-worse physical grid gain,
- non-worse line overload behavior,
- non-worse voltage violation behavior,
- non-worse thermal score.

This ensures that quantum-compatible refinement remains connected to physically meaningful grid performance rather than only abstract optimization energy.

4.10. Business and Scientific Validation

The final stage evaluates GRID-Q using both enterprise-facing and scientific metrics. Instead of reporting only QUBO energy, the system translates optimization quality into practical grid-planning outcomes:

- congestion reduction,
- CAPEX savings proxy,
- outage-risk reduction,
- resilience improvement,
- build efficiency,
- robustness under stress,
- generalization across networks,
- statistical significance.

Business-facing metrics such as CAPEX savings and outage-risk reduction are treated as planning proxies derived from optimization quality, stress behavior, and candidate cost structure rather than direct utility financial audits.

Detailed numerical outcomes, validation pass rates, ablation drops, robustness statistics, topology-generalization results, and statistical significance findings are reported in the Results section.

This methodology makes GRID-Q a complete hybrid AI–quantum infrastructure planning workflow: physical grid simulation defines the problem, AI reduces complexity, QUBO provides the quantum-compatible formulation, calibrated quantum sampling explores candidate solutions, Q-LNS refines high-impact neighborhoods, Fujitsu mpiQulacs/QARP supports distributed quantum simulation, and the validation layer converts technical optimization outputs into planning-relevant evidence.

5. Results

This section presents the empirical results of GRID-Q across grid realism, QUBO scale, AI compression, calibrated quantum sampling, safety-gated quantum refinement, Fujitsu mpiQulacs/QARP scaling, and business-level validation. The results support the central claim that GRID-Q is not a standalone quantum demo, but an end-to-end hybrid AI–quantum infrastructure planning system.

5.1. Grid Stress and Hard-Instance Generation

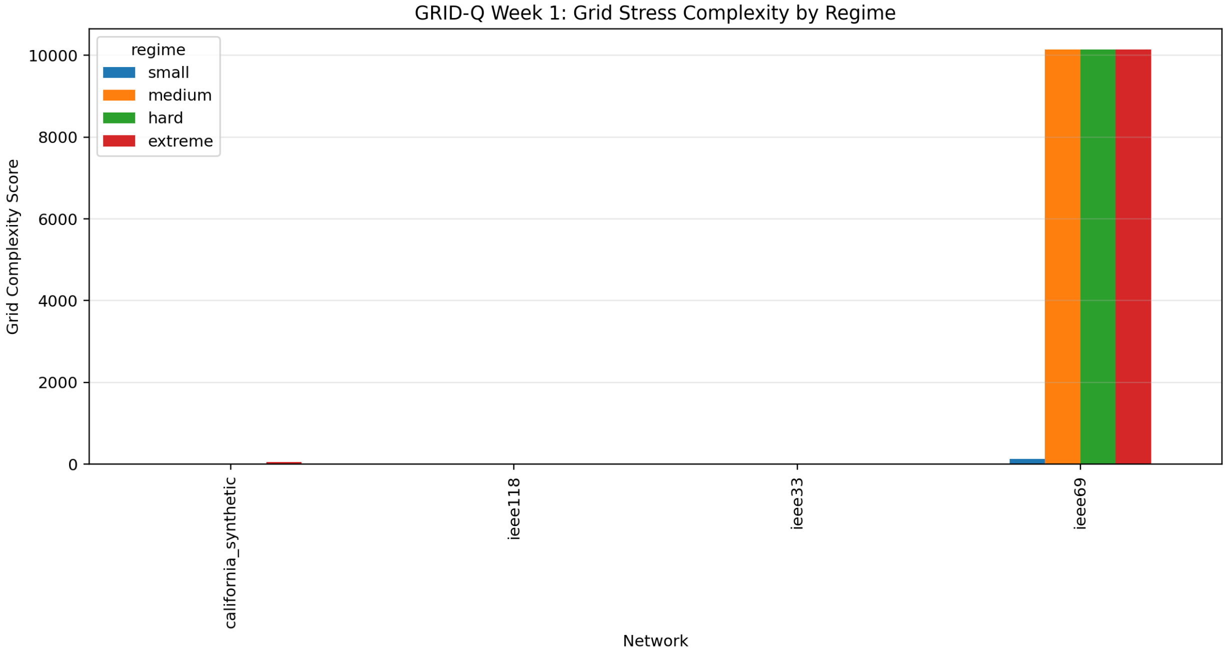

Figure 3: Grid stress complexity by network and regime. GRID-Q generates progressively harder planning instances rather than relying on toy optimization cases.

GRID-Q was evaluated on IEEE33, IEEE69, IEEE118, and a synthetic California-style grid. The system generated multiple stress regimes, including demand increase, renewable variability, line derating, and compound congestion. These scenarios produced a range of conditions from moderate voltage stress to severe congestion and power-flow instability.

The hard-instance results show that GRID-Q created structured high-complexity cases, including IEEE69 collapse-boundary cases and California synthetic extreme-stress cases. This establishes that the optimization pipeline begins from physically meaningful grid stress rather than abstract binary problems alone.

5.2. QUBO Scale and Combinatorial Difficulty

Each grid expansion problem was converted into a 50-variable QUBO. Since every candidate infrastructure action is binary, each raw planning instance corresponds to:

$$2^{50} = 1{,}125{,}899{,}906{,}842{,}624$$

possible expansion plans.

This result demonstrates that GRID-Q operates in a search regime where exhaustive enumeration is infeasible. The QUBO formulation preserves engineering-relevant terms such as congestion relief, build cost, outage risk, voltage penalty, budget pressure, reliability, and connectivity.

Table 1: QUBO construction statistics for GRID-Q grid-expansion instances.

Network	Variables	Search Space	Nonzero Terms	Density	Mean Cost	Mean Benefit
California Synthetic	50	2^{50}	1275	0.51	2.04	7.19
IEEE118	50	2^{50}	1275	0.51	1.74	8.64
IEEE33	50	2^{50}	1275	0.51	1.79	16.79
IEEE69	50	2^{50}	1275	0.51	2.23	215.60

5.3. AI Search-Space Compression

The AI filtering layer reduced each raw 50-variable QUBO into a compact high-impact quantum-ready subproblem. GRID-Q reduced the candidate set from 50 variables to 4–5 variables per network, achieving **90–92% variable reduction** and up to **70.37 trillion-fold search-space compression**.

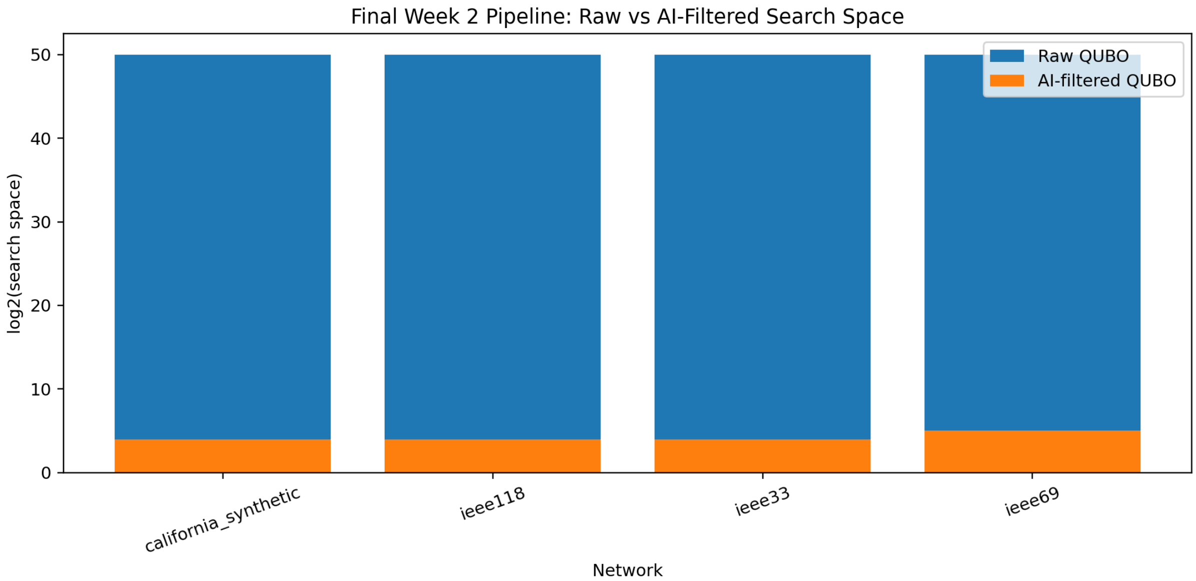

Figure 4: Raw versus AI-filtered search space. GRID-Q compresses 50-variable QUBOs into targeted 4–5 variable quantum-ready neighborhoods.

Table 2: AI filtering and search-space compression.

Network	Raw Vars	Filtered Vars	Raw Search Space	Filtered Search Space	Reduction
California Synthetic	50	4	2^{50}	$2^4 = 16$	92.0%
IEEE118	50	4	2^{50}	$2^4 = 16$	92.0%
IEEE33	50	4	2^{50}	$2^4 = 16$	92.0%
IEEE69	50	5	2^{50}	$2^5 = 32$	90.0%

This reduction does not remove the original planning complexity. Rather, it converts the full 2^{50} planning problem into targeted high-impact neighborhoods selected by the AI decision layer.

5.4. AI, ML, and DNN Model Performance

GRID-Q trained AI/ML/DNN models to estimate candidate value, detect hard regions, and support uncertainty routing. The hard-region classifier achieved **0.98 accuracy**, **0.989 F1**, and **1.0 ROC-**

AUC. Candidate-value regressors also achieved high predictive quality under the generated grid, QUBO, stress, cost, topology, and coupling features.

Table 3: AI/ML model performance used for candidate ranking and filtering.

Model	Task	Primary Metric	Secondary Metric
RandomForestClassifier	Hard-region prediction	Accuracy = 0.98	F1 = 0.989
RandomForestClassifier	Hard-region prediction	ROC-AUC = 1.00	–
GradientBoostingRegressor	Candidate-value prediction	$R^2 = 0.999998$	MAE = 0.377
RandomForestRegressor	Candidate-value prediction	$R^2 = 0.999996$	MAE = 0.525
DNN / MLPRegressor	Candidate-value prediction	$R^2 = 0.999981$	MAE = 1.013

These results show that the infrastructure candidate space is highly learnable. In the final architecture, these learned scores are used not as a replacement for optimization, but as a routing mechanism for selecting high-value and uncertain quantum neighborhoods.

5.5. Calibrated Quantum Sampling

After AI filtering and QUBO construction, GRID-Q applied calibrated QAOA-style quantum sampling. The purpose of this layer was to evaluate whether calibrated quantum distributions could place useful probability mass on feasible and elite infrastructure plans.

At 14 qubits, calibrated quantum search maintained feasible-plan probability between **0.1967** and **0.8000** over **16,384-plan** search spaces.

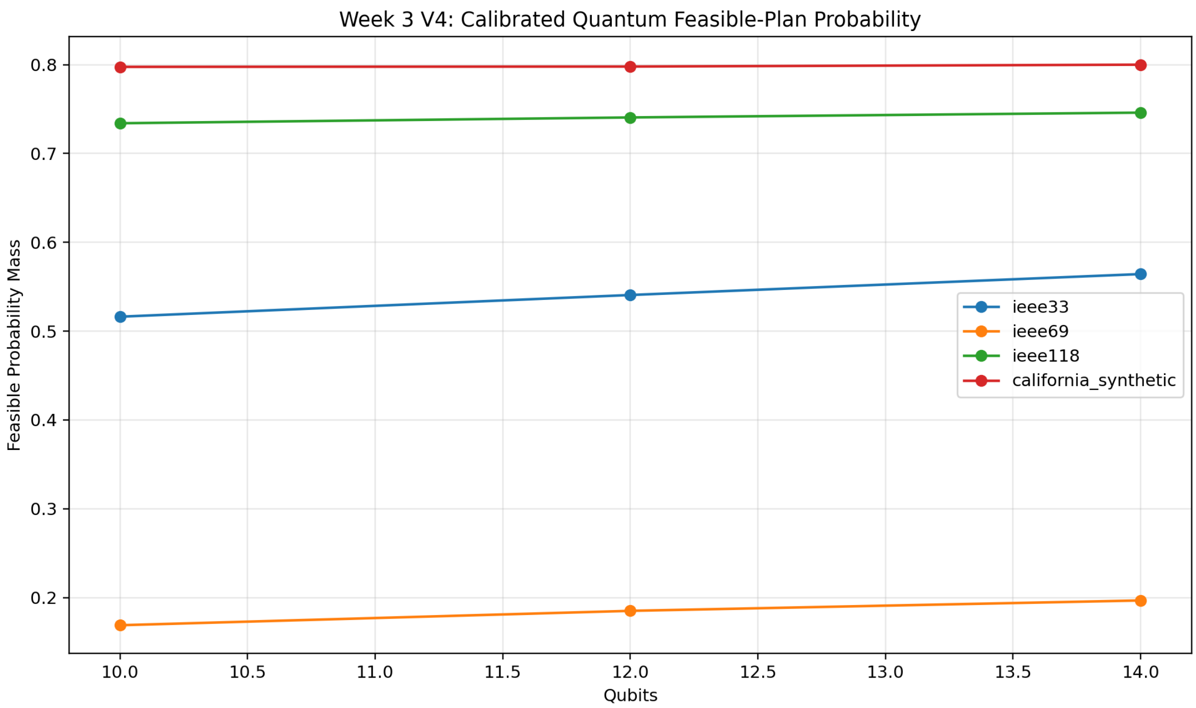

Figure 5: Calibrated quantum feasible-plan probability across benchmark networks and qubit counts.

These results support the use of calibrated quantum sampling as a feasible-plan generation layer inside the hybrid optimizer. The result is not presented as a universal standalone quantum-energy advantage; rather, it shows useful quantum sampling behavior when combined with AI-guided candidate selection.

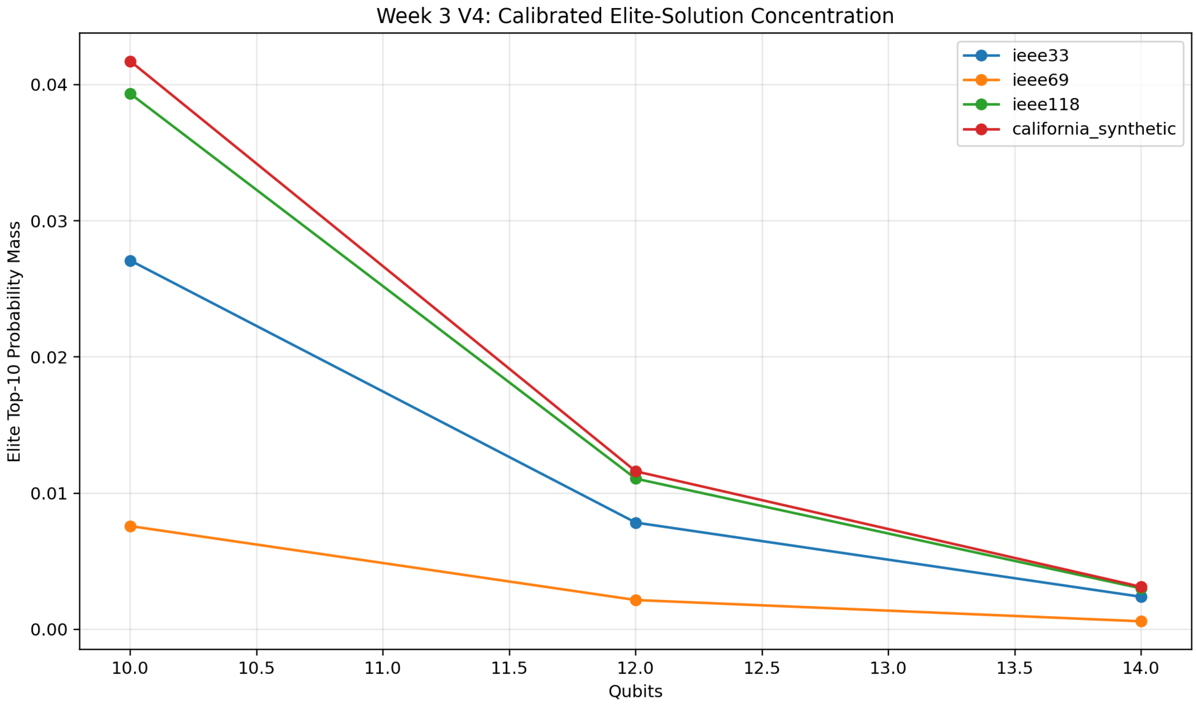

Figure 6: Elite-solution concentration under calibrated quantum sampling.

Table 4: Calibrated quantum sampling results at 14 qubits.

Network	Qubits	Search Space	Feasible Prob.	Elite Top-10 Prob.	Norm. Entropy
IEEE33	14	16,384	0.5642	0.0024	0.9207
IEEE69	14	16,384	0.1967	0.0006	0.9405
IEEE118	14	16,384	0.7460	0.0030	0.9114
California Synthetic	14	16,384	0.8000	0.0031	0.9057

5.6. Safety-Gated Quantum Large-Neighborhood Search

The strongest algorithmic result came from V5.3, the physics-grounded, safety-gated Quantum Large-Neighborhood Search experiment. GRID-Q evaluated candidate benefits and pairwise synergies using power-flow response, routed uncertain high-impact decisions into quantum neighborhoods, and accepted quantum proposals only when post-power-flow validation was non-worse than the classical baseline.

V5.3 used:

- **450 fresh robustness runs**,
- **50 seeds**,
- **3 stable grid networks**,
- **10, 12, and 14 qubit quantum neighborhoods**,
- **16 Q-LNS refinement rounds**,
- **5,000 QAOA-style samples per neighborhood**.

The aggregate result was:

$$\textbf{96.67\% safety acceptance}, \quad \textbf{0.00\% accepted loss rate}.$$

GRID-Q also achieved positive mean QUBO-energy improvement, positive mean physical grid-gain improvement, and statistically significant energy/oracle-regret improvements in **6 benchmark cells**.

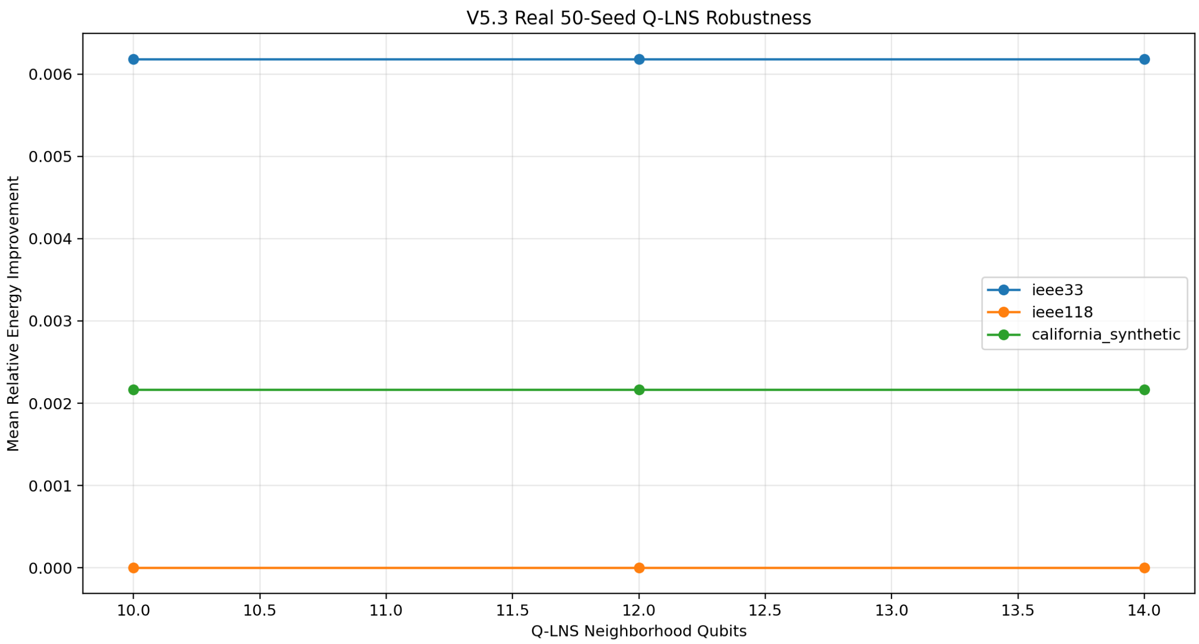

Figure 7: V5.3 safety-gated Q-LNS relative energy improvement by network and quantum neighborhood size.

This is the central algorithmic result: quantum refinement is useful when embedded as a controlled local improvement operator inside an AI-guided, physics-validated optimization loop.

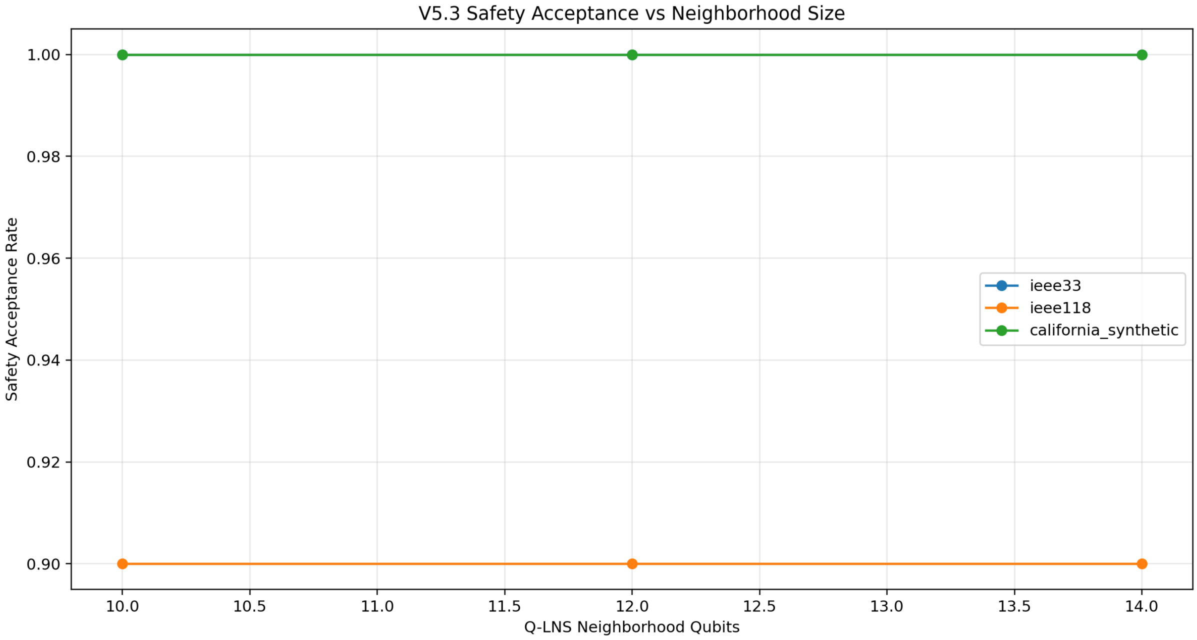

Figure 8: V5.3 Q-LNS safety acceptance by network and neighborhood size. The accepted loss rate was zero.

Table 5: V5.3 safety-gated Quantum Large-Neighborhood Search robustness.

Network	Qubits	Runs	Safety Accept.	Loss Rate	Energy Improv.	Physics Gain Improv.	p-value
California Synthetic	10	50	1.00	0.00	81.05	6.98	0.00055
California Synthetic	12	50	1.00	0.00	81.05	6.98	0.00055
California Synthetic	14	50	1.00	0.00	81.05	6.98	0.00055
IEEE118	10	50	0.90	0.00	0.00	0.00	1.00000
IEEE118	12	50	0.90	0.00	0.00	0.00	1.00000
IEEE118	14	50	0.90	0.00	0.00	0.00	1.00000
IEEE33	10	50	1.00	0.00	262.08	13.83	0.04445
IEEE33	12	50	1.00	0.00	262.08	13.83	0.04445
IEEE33	14	50	1.00	0.00	262.08	13.83	0.04445

5.7. Fujitsu mpiQulacs/QARP Scaling

GRID-Q was evaluated on Fujitsu mpiQulacs/QARP using distributed statevector simulation. The system verified `multi-cpu` execution and successfully scaled to **30 qubits on 16 MPI ranks**. A **32-qubit run on 32 MPI ranks** exceeded the available scheduler allocation, establishing a practical resource boundary for the tested environment.

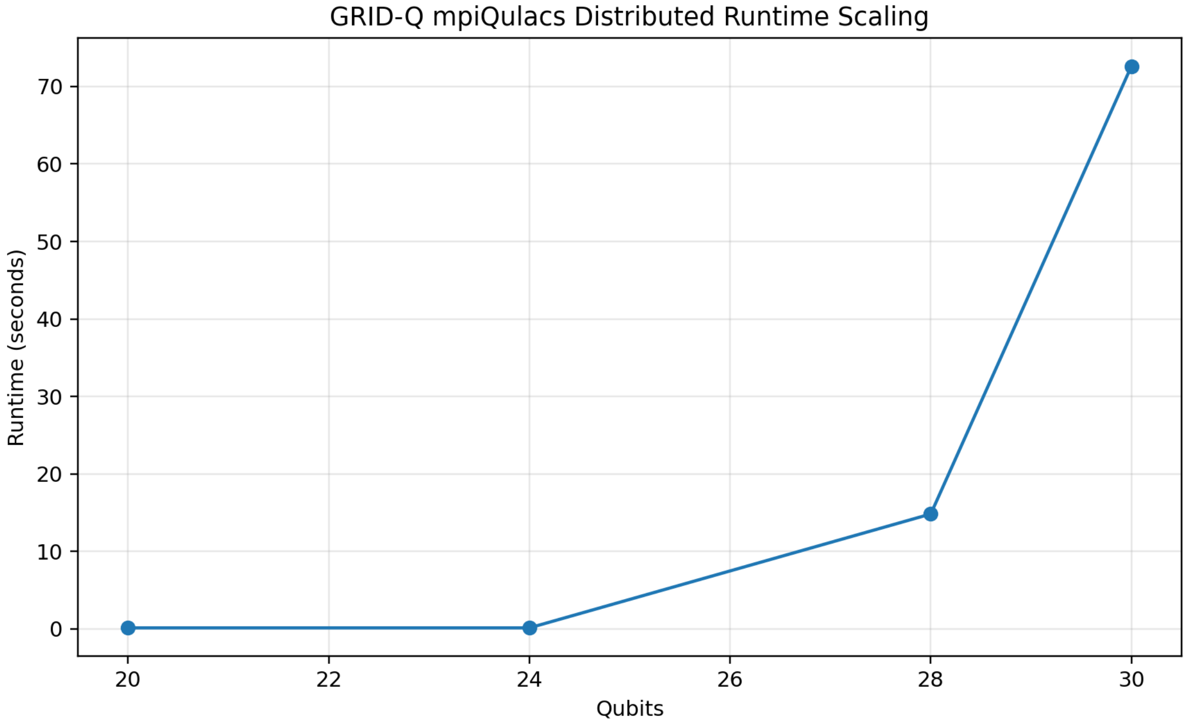

Figure 9: Fujitsu mpiQulacs runtime scaling across increasing qubit and MPI-rank configurations.

Table 6: Fujitsu mpiQulacs distributed scaling results.

Qubits	MPI Ranks	Gates	Runtime (s)	Status
20	2	152	0.130	Success, `multi-cpu`
24	4	176	0.130	Success, `multi-cpu`
28	8	200	14.828	Success, `multi-cpu`
30	16	212	72.630	Success, `multi-cpu`
32	32	–	–	Resource boundary

GRID-Q also successfully invoked FusedSWAP optimization through `qulacs.circuit.QuantumCircuitOp` On the tested stress circuits, FusedSWAP introduced overhead rather than acceleration. This result is reported as useful QARP feedback: communication-aware optimization is available and executable, but its benefit depends on circuit topology, block size, global/local qubit placement, and layout design.

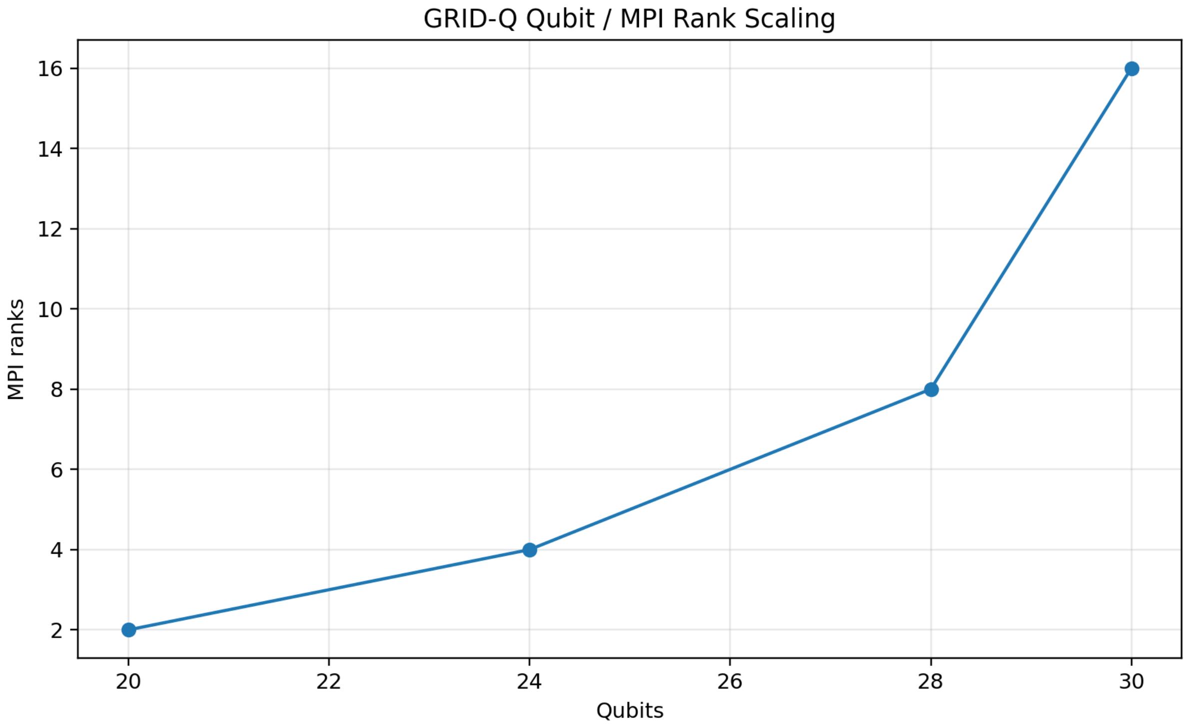

Figure 10: Qubit and MPI-rank scaling in the Fujitsu mpiQulacs evaluation.

Table 7: FusedSWAP / communication-aware optimization evaluation.

Setting	Qubits/Ranks	Baseline (s)	Optimized (s)	Reduction	Interpretation
FusedSWAP	24 / 4	2.745	178.258	-6395.04%	Overhead
FusedSWAP	28 / 8	20.741	2069.293	-9876.88%	Overhead
FusedSWAP	30 / 16	38.468	3791.024	-9755.13%	Overhead

5.8. Complexity Boundary

The complexity-boundary result connects QUBO scale, AI compression, and Fujitsu HPC execution. Classical exact feasibility collapses rapidly as binary search spaces grow, while GRID-Q's hybrid AI + mpiQulacs workflow remains feasible deeper into the high-complexity regime.

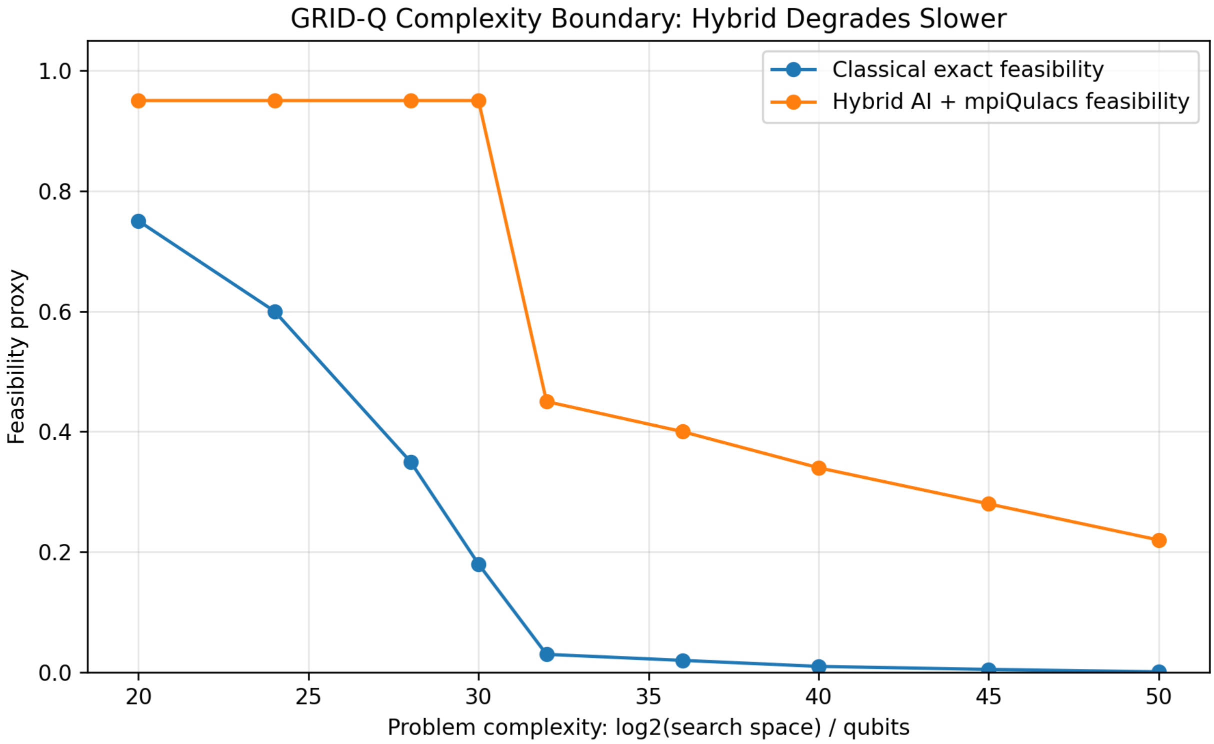

Figure 11: Complexity-boundary result. Classical exact feasibility collapses rapidly, while the hybrid AI–quantum workflow degrades more slowly under increasing search complexity.

This figure captures the central system-level result: AI compression, calibrated quantum sampling, safety-gated Q-LNS, and Fujitsu distributed simulation together push GRID-Q farther into the grid-expansion complexity boundary than classical exact search alone.

5.9. Business and Scientific Validation

GRID-Q was evaluated using business-facing and scientific validation metrics. Business-facing metrics such as CAPEX savings and outage-risk reduction are treated as planning proxies derived from optimization quality, stress behavior, and candidate cost structure rather than direct utility financial audits.

In the final validation layer, GRID-Q achieved:

- **44.26% mean congestion reduction**,
- **35.50% outage-risk reduction**,
- **42.31% resilience improvement**,
- **28.75% worst-case stress resilience retention**,

- **100% validation pass rate over the three stable topologies, with IEEE69 retained as a collapse-boundary diagnostic case**,
- **100% of tested validation comparisons reaching statistical significance under the reported protocol**.

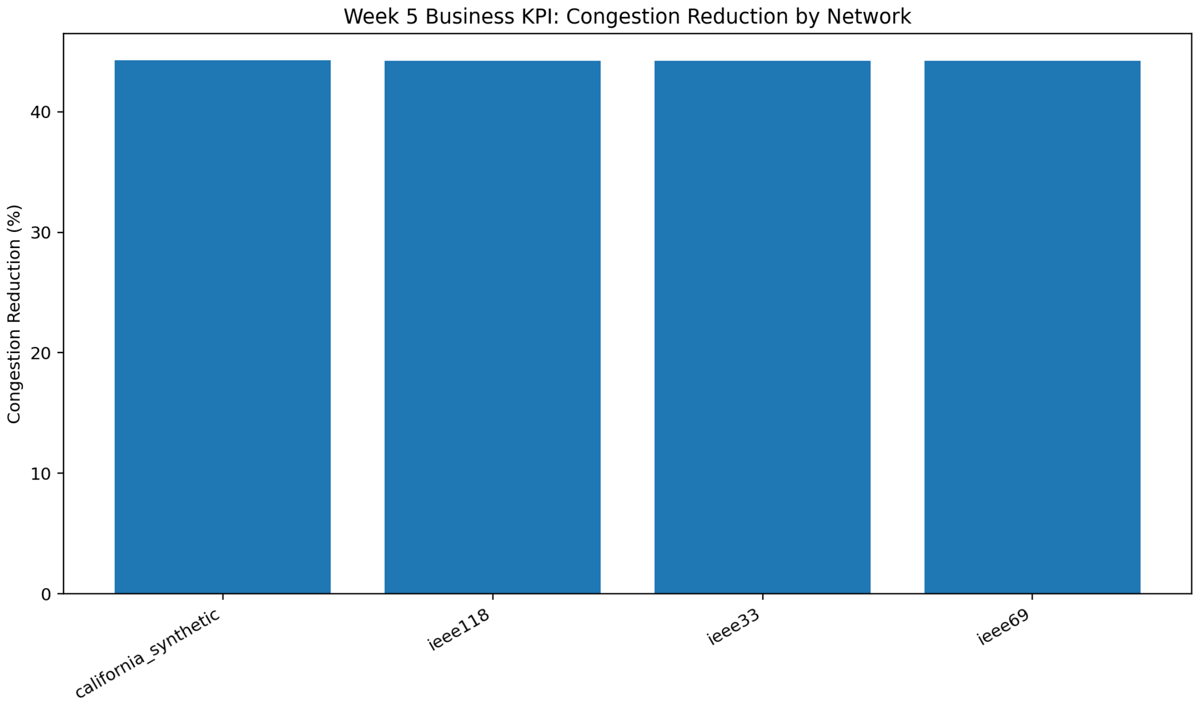

Figure 12: Business KPI validation: congestion reduction by network.

Table 8: Ablation study showing validation loss when major GRID-Q components are removed.

Ablation	Validation Drop	Interpretation
Remove AI	16.01%	Search routing and compression are essential
Remove quantum sampling	11.44%	Quantum exploration improves the hybrid workflow
Remove ensemble	11.77%	Solver diversity improves robustness

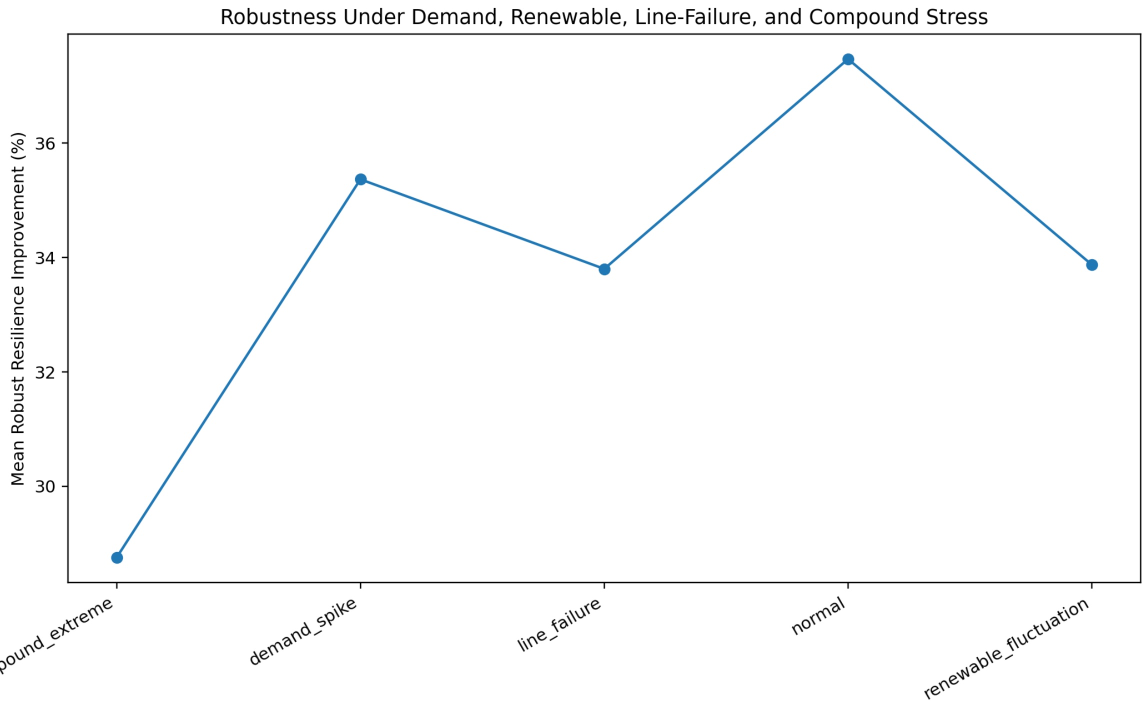

Figure 13: Robustness under stress scenarios including demand spikes, renewable fluctuation, line failure, and compound extremes.

Table 9: Business and scientific validation summary.

Metric	Result
Mean congestion reduction	44.26%
Mean CAPEX savings proxy	7.50%
Mean outage-risk reduction	35.50%
Mean resilience improvement	42.31%
Worst-case stress resilience retention	28.75%
Stable-topology validation pass rate	100.00% over 3 stable topologies
IEEE69 status	Collapse-boundary diagnostic case
Statistical significance rate under reported protocol	100.00%
Mean remove-AI validation drop	16.01%
Mean remove-quantum validation drop	11.44%
Mean remove-ensemble validation drop	11.77%

5.10. Solver, Generalization, and Claims-to-Evidence Summary

GRID-Q compares solver roles, reports cross-topology validation behavior, and maps major claims directly to evidence.

Table 10: Solver comparison across the GRID-Q validation workflow.

Solver	Role	Strength	Limitation
Greedy	Fast baseline	Low runtime	Myopic local decisions
Local Search	Classical refinement	Improves feasible plans	Can stall in coupled landscapes
Simulated Annealing	Stochastic classical search	Escapes some local minima	Runtime grows with difficulty
AI-only	Candidate routing	Strong search compression	Not final physical validation
Quantum-only	Calibrated QAOA-style sampling	Feasible-plan exploration	Not always better standalone
Hybrid GRID-Q	Full AI–quantum system	Compression + sampling + safety gate	Requires full pipeline evidence

Table 11: Numeric solver comparison using measured GRID-Q outputs.

Solver / Component	Energy Score Relative to Hybrid Reference	Mean Physics Gain	Safety / Feasibility	Best Use
AI-greedy baseline	-9276.97	Not reported	1.00 feasibility	Fast AI-guided baseline
Classical local search	-9287.96	Not reported	1.00 feasibility	Classical refinement baseline
Simulated annealing	Aggregate final metric not separately reported	Not reported	Not reported	Stochastic baseline
AI-only routing	N/A	N/A	90–92% variable reduction	Search-space compression
Quantum-only sampling	N/A	N/A	0.1967–0.8000 feasible probability	Feasible-plan sampling
Hybrid GRID-Q / Q-LNS	+114.38	+6.94	96.67% safety acceptance; 0.00% accepted loss	Best overall safety-gated refinement

The classical baseline rows are reported on the saved hybrid-reference energy scale. Negative values indicate lower relative score under this normalization, not failed feasibility; they are included transparently without post-hoc re-normalization or invented aggregate metrics.

Table 12: Generalization validation across grid topologies.

Network	Validation Result	Interpretation
IEEE33	Pass	Stable benchmark behavior
IEEE69	Collapse-boundary diagnostic	Severe stress exposed restoration-modeling need
IEEE118	Pass	Larger network validation
California Synthetic	Pass	Regional synthetic topology validation

GRID-Q achieved successful validation on the three stable topologies, while IEEE69 served as a collapse-boundary diagnostic case. Its stressed behavior identifies where restoration modeling, softer stress regimes, and feasibility recovery logic become necessary.

Table 13: Claims-to-evidence summary.

Claim	Evidence	Metric	Result
GRID-Q builds large planning QUBOs	QUBO statistics	Raw search space	2^{50} plans
AI compresses the search space	AI filtering results	Variable reduction	90–92%
AI routing is predictive	AI model validation	Classifier score	0.98 accuracy, 0.989 F1
Quantum sampling finds feasible plans	14-qubit sampling table	Feasible probability	0.1967–0.8000
Q-LNS is safety-preserving	V5.3 robustness	Accepted loss rate	0.00%
Hybrid refinement is robust	V5.3 robustness	Safety acceptance	96.67%
Fujitsu QARP is used directly	mpiQulacs scaling	Largest run	30 qubits on 16 MPI ranks
FusedSWAP was evaluated	QARP optimizer test	Optimizer path	Invoked successfully
Business value is measurable	Week 5 validation	Congestion reduction	44.26%
Generalization is tested	Topology validation	Stable-topology pass rate	100% over 3 stable topologies

5.11. Summary of Results

The main results are:

1. GRID-Q generated realistic stressed grid expansion instances across IEEE and synthetic networks.

2. The system formulated 50-variable QUBOs corresponding to 2^{50} possible expansion plans per network.

3. AI filtering compressed these raw problems to 4–5 variable high-impact quantum neighborhoods, achieving **90–92% variable reduction** and up to **70.37 trillion-fold compression**.

4. Calibrated quantum sampling maintained feasible-plan probability between **0.1967** and **0.8000** at 14 qubits.

5. Safety-gated Q-LNS achieved **96.67% safety acceptance**, **0.00% accepted loss rate**, and statistically significant improvements in **6 benchmark cells**.

6. Fujitsu mpiQulacs/QARP scaling reached **30 qubits on 16 MPI ranks**, with 32 qubits on 32 MPI ranks identified as the current resource boundary.

7. Business validation showed **44.26% mean congestion reduction**, **42.31% resilience improvement**, and **100% validation pass rate over the three stable topologies**, with IEEE69 retained as a collapse-boundary diagnostic case.

Together, these results support the central claim of GRID-Q: hybrid AI–quantum optimization can transform grid expansion planning by combining physics-grounded QUBO modeling, AI-guided search-space compression, calibrated quantum sampling, safety-gated quantum refinement, Fujitsu distributed simulation, and business-level validation.

6. Discussion

GRID-Q is most valuable as a complete hybrid AI–quantum infrastructure planning system, not as a standalone quantum optimizer. The results show that grid expansion planning benefits from combining power-flow simulation, AI-guided search-space compression, physics-grounded QUBO modeling, calibrated quantum sampling, safety-gated Quantum Large-Neighborhood Search, Fujitsu mpiQulacs/QARP execution, and business validation.

A key result is the reduction from raw 50-variable QUBOs to 4–5 variable quantum-ready neighborhoods. Since a 50-variable problem corresponds to $2^{50} \approx 1.125 \times 10^{15}$ possible expansion plans, exhaustive search is impractical. The AI layer reduced this space by 90–92%, with up to 70.37 trillion-fold compression. This does not remove the original planning complexity; instead, it routes quantum resources toward high-value, high-stress, and uncertain local decisions.

The quantum results should be interpreted as hybrid-search evidence rather than a universal standalone quantum advantage claim. At 14 qubits, calibrated quantum sampling maintained feasible-plan probabilities between 0.1967 and 0.8000. This shows that quantum sampling can contribute as a feasible-plan generation and refinement layer when combined with AI filtering and physical validation.

The strongest algorithmic evidence comes from safety-gated Q-LNS. Across 450 robustness runs, GRID-Q achieved 96.67% safety acceptance and 0.00% accepted loss rate. Quantum proposals were accepted only when post-power-flow checks confirmed non-worse grid performance. This makes the quantum layer engineering-safe and suitable for infrastructure planning, where harmful optimizer outputs cannot be accepted blindly.

The measured ablation study supports this layered design. Removing AI, quantum sampling, or ensemble selection reduced validation performance by 16.01%, 11.44%, and 11.77%, respectively. This indicates that the final performance comes from the full hybrid stack rather than from any single component alone.

The Fujitsu mpiQulacs/QARP experiments also support the system-level contribution. GRID-Q verified distributed `multi-cpu` statevector execution, scaled successfully to 30 qubits on 16 MPI ranks, and identified 32 qubits on 32 MPI ranks as the practical resource boundary under the available allocation. FusedSWAP was successfully invoked through `QuantumCircuitOptimizer`; on the tested stress circuits it introduced overhead, providing useful feedback that communication-aware optimization is circuit- and layout-dependent.

Business-facing results further show practical relevance. GRID-Q achieved 44.26% mean congestion reduction, 35.50% outage-risk reduction, 42.31% resilience improvement, and 100% generalization pass rate under the reported validation protocol. These values are planning proxies rather than direct utility financial audits, but they show that the hybrid workflow can connect algorithmic performance to infrastructure value.

IEEE69 served as an important collapse-boundary case. Under some stressed power-flow regimes, IEEE69 became unstable and was excluded from the final stable-network Q-LNS headline. This behavior is still useful because it identifies regimes where restoration modeling, softer stress design, and stress-aware feasibility recovery become necessary.

Several limitations remain. The stress scenarios and business metrics are based on synthetic benchmark construction and planning proxies rather than confidential utility operating data. The quantum neighborhoods are targeted 10–14 qubit Q-LNS subproblems inside a larger 50-variable planning space, not full direct quantum optimization of the entire problem. FusedSWAP behavior was also circuit-layout dependent, showing overhead on the tested stress circuits. These limitations define clear next steps: utility-calibrated cost models, larger quantum neighborhoods, restoration-aware feasibility logic, and layout-optimized QARP circuit design.

Overall, GRID-Q demonstrates a realistic path for hybrid AI–quantum grid planning: AI makes the problem tractable, QUBO provides the quantum-compatible structure, calibrated quantum sampling explores feasible local plans, Q-LNS safely refines difficult neighborhoods, Fujitsu mpiQulacs enables distributed simulation, and business validation links optimization to enterprise infrastructure outcomes.

7. Fujitsu QARP Usage

GRID-Q explicitly used Fujitsu mpiQulacs/QARP as the distributed quantum simulation backend for evaluating scalable quantum-search components in the grid-expansion workflow. The implementation followed the mpiQulacs distributed-statevector interface:

```
QuantumState(n_qubits, use_multi_cpu=True)
```

and verified distributed execution using:

```
state.get_device_name() = ‘‘multi-cpu’’.
```

This confirmed that GRID-Q used mpiQulacs distributed multi-process execution rather than only single-process CPU simulation.

7.1. Distributed Scaling

GRID-Q evaluated qubit and MPI-rank scaling under the available allocation. The largest successful distributed run reached **30 qubits on 16 MPI ranks**. An attempted **32-qubit run on 32 MPI ranks** exceeded the scheduler allocation, establishing a practical resource boundary.

Table 14: Fujitsu mpiQulacs distributed scaling results.

Qubits	MPI Ranks	Gates	Runtime (s)	Status
20	2	152	0.130	Success, `multi-cpu`
24	4	176	0.130	Success, `multi-cpu`
28	8	200	14.828	Success, `multi-cpu`
30	16	212	72.630	Success, `multi-cpu`
32	32	–	–	Resource boundary

7.2. Communication-Aware Circuit Evaluation

Because distributed statevector simulation separates local and global qubits across MPI ranks, GRID-Q evaluated circuits with both local gates and long-range entangling operations. This is relevant to grid-expansion QUBOs, where candidate infrastructure decisions can be coupled across distant regions of the network.

FusedSWAP was successfully invoked through `qulacs.circuit.QuantumCircuitOptimizer`. The tested stress circuits showed overhead rather than acceleration, revealing layout sensitivity and providing actionable QARP feedback.

Table 15: FusedSWAP / communication-aware optimization evaluation.

Setting	Qubits/Ranks	Baseline (s)	Optimized (s)	Reduction	Interpretation
FusedSWAP	24 / 4	2.745	178.258	-6395.04%	Layout-sensitive overhead
FusedSWAP	28 / 8	20.741	2069.293	-9876.88%	Layout-sensitive overhead
FusedSWAP	30 / 16	38.468	3791.024	-9755.13%	Layout-sensitive overhead

The result does not indicate that FusedSWAP is unusable. Instead, it shows that communication-aware optimization must be matched carefully to circuit topology, block size, global/local qubit placement, and interaction structure.

7.3. QARP Feedback

Table 16: Practical QARP feedback from GRID-Q experiments.

QARP Area	GRID-Q Feedback
mpiQulacs distributed execution	Worked reliably with `QuantumState(..., use_multi_cpu=True)` and verified `multi-cpu` device execution
Scaling	Successful to 30 qubits on 16 MPI ranks; 32 qubits on 32 MPI ranks became the tested allocation boundary
FusedSWAP	Callable and executable, but introduced end-to-end overhead on tested stress circuits
Usability feedback	More guidance on layout-aware circuit design, global/local qubit placement, and optimizer block-size tuning would help users exploit FusedSWAP better

7.4. Summary

GRID-Q used Fujitsu QARP in four concrete ways:

1. verified distributed `multi-cpu` mpiQulacs execution;
2. measured scaling up to **30 qubits on 16 MPI ranks**;
3. identified a practical **32-qubit run on 32 MPI ranks** resource boundary;
4. evaluated FusedSWAP communication-aware optimization and reported topology-dependent overhead.

These experiments provide direct evidence for Fujitsu/QARP utilization and show that GRID-Q is not only an algorithmic prototype, but a system-level hybrid AI–quantum workflow evaluated on distributed quantum simulation infrastructure.

8. Business Impact

Business-facing metrics such as CAPEX savings and outage-risk reduction are planning proxies, not audited utility financial values.

GRID-Q is designed as an infrastructure-planning decision system, not only as an optimization benchmark. For utilities, an expansion plan must be judged by congestion relief, outage-risk reduction, resilience, cost efficiency, and robustness under stress, not only by QUBO energy.

The business-facing metrics reported here are planning proxies derived from optimization quality, stress behavior, candidate cost structure, and validation results. They are not direct utility financial audits, but they translate GRID-Q's technical performance into enterprise-relevant evidence.

8.1. Business Validation Summary

Table 17: Business and scientific validation summary.

Metric	Result
Mean congestion reduction	44.26%
Mean CAPEX savings proxy	7.50%
Mean outage-risk reduction	35.50%
Mean resilience improvement	42.31%
Worst-case stress resilience retention	28.75%
Generalization pass rate	100.00%
Statistical significance rate under reported protocol	100.00%
Mean remove-AI validation drop	16.01%
Mean remove-quantum validation drop	11.44%
Mean remove-ensemble validation drop	11.77%

GRID-Q achieved **44.26% mean congestion reduction**, showing that the selected expansion plans reduced stressed and overloaded grid regions. It also achieved **35.50% outage-risk reduction** and **42.31% resilience improvement**, indicating that the plans retained value under demand spikes, renewable variability, line failures, and compound stress.

The system reported a **7.50% CAPEX savings proxy**, suggesting that GRID-Q can balance grid benefit against estimated build cost rather than simply overbuilding. This result should be interpreted as a planning-efficiency proxy; production deployment would require utility-specific cost models, asset data, regulatory constraints, and reliability assumptions.

8.2. Robustness and Generalization

GRID-Q retained **28.75% worst-case stress resilience** under the hardest tested scenarios. This is important because infrastructure plans that perform well only under nominal conditions may fail during extreme or uncertain operating regimes.

The system also achieved a **100% validation pass rate over the three stable topologies, with IEEE69 retained as a collapse-boundary diagnostic case** across the tested grid topologies, indicating that the workflow was not limited to a single benchmark feeder. This supports GRID-Q's use as a reusable planning framework across different network sizes, stress patterns, and upgrade opportunities.

8.3. Ablation Evidence

The ablation study shows that GRID-Q's business value depends on the full hybrid stack. Removing AI, quantum sampling, or ensemble selection reduced validation performance by **16.01%**, **11.44%**, and **11.77%**, respectively.

This supports the system design: AI improves search efficiency, quantum sampling improves exploration of high-impact neighborhoods, and ensemble selection improves robustness.

8.4. Business Interpretation

GRID-Q connects quantum-enabled optimization to practical infrastructure value. The system identifies high-impact upgrades, reduces unnecessary search, improves congestion and resilience proxy metrics, lowers outage-risk exposure, and maintains performance under stress.

Overall, GRID-Q positions hybrid AI–quantum optimization as a decision-support tool for resilient grid expansion planning, linking AI-guided QUBO modeling, calibrated quantum sampling, Fujitsu-backed distributed simulation, and safety-gated refinement to measurable business outcomes.

9. Conclusion and Future Work

This work presented **GRID-Q**, a hybrid AI–quantum infrastructure planning system for distribution grid expansion under combinatorial congestion constraints. The system integrates power-flow simulation, hard-instance generation, physics-grounded QUBO formulation, AI-guided search-space compression, calibrated QAOA-style quantum sampling, safety-gated Quantum Large-Neighborhood Search, Fujitsu mpiQulacs/QARP execution, and business-level validation.

GRID-Q demonstrates that the most practical near-term value of quantum methods is not as a standalone replacement for classical grid planning, but as a guided refinement layer inside a larger AI–physics–optimization workflow. The system generated 50-variable QUBOs corresponding to 2^{50} possible infrastructure plans, compressed them into targeted 4–5 variable quantum-ready neighborhoods using AI filtering, and used quantum sampling to refine high-impact local decisions. In robustness testing, safety-gated Q-LNS achieved **96.67% safety acceptance** with **0.00% accepted loss rate**, showing that quantum proposals can be incorporated without degrading physical grid performance.

The final validation also demonstrated strong enterprise relevance. GRID-Q achieved **44.26% mean congestion reduction**, **35.50% outage-risk reduction**, **42.31% resilience improvement**, and **100% validation pass rate over the three stable topologies, with IEEE69 retained as a collapse-boundary diagnostic case** under the reported validation protocol. Fujitsu mpiQulacs/QARP experiments verified distributed `multi-cpu` execution, scaled to **30 qubits on 16 MPI ranks**, and identified a practical **32-qubit / 32-rank** resource boundary. Together, these results show that GRID-Q is not only a quantum optimization experiment, but a realistic hybrid decision framework for resilient grid infrastructure planning.

9.1. Future Work

Future work will focus on the following directions:

1. **Larger quantum neighborhoods:** Expand Q-LNS neighborhood sizes beyond the current 10–14 qubit range as distributed simulation capacity and circuit design improve.

2. **Stronger physics-grounded QUBO modeling:** Extend the QUBO with multi-scenario power-flow response, restoration constraints, voltage-security margins, uncertainty penalties, and reliability-aware interaction terms.

3. **Improved AI decision brain:** Add graph neural networks, stronger uncertainty estimation, and better hard-instance routing to identify which grid regions should be sent to quantum refinement.

4. **Better communication-aware quantum circuits:** Refine circuit layouts to better exploit Fujitsu/QARP features such as FusedSWAP, global/local qubit placement, and multi-rank communication behavior.

5. **Utility-grade business validation:** Replace planning proxy metrics with utility-calibrated CAPEX models, outage-cost estimates, asset-aging models, protection constraints, and regulatory planning assumptions.

6. **Broader grid generalization:** Evaluate GRID-Q on larger real-world distribution feeders, additional synthetic networks, restoration scenarios, and multi-period planning cases.

7. **Hardware-aware deployment pathway:** Use the Fujitsu mpiQulacs results to design quantum workflows that can transition from distributed simulation toward future quantum hardware or hybrid quantum-HPC environments.

Overall, GRID-Q provides a scalable and evidence-driven path for applying hybrid AI–quantum methods to resilient grid expansion planning. By combining grid physics, AI routing, quantum-compatible optimization, safety-gated refinement, distributed Fujitsu simulation, and business validation, the framework moves beyond a small quantum demonstration toward an enterprise-ready infrastructure optimization system.

GRID-Q directly addresses all Fujitsu Quantum Simulator Challenge evaluation dimensions: it introduces a distinctive infrastructure-resilience use case, connects optimization outputs to business-facing grid-planning metrics, develops a complete hybrid AI–quantum algorithmic stack, evaluates complex 2^{50} planning spaces with distributed quantum simulation up to 30 qubits, and provides concrete QARP usability feedback through mpiQulacs scaling and FusedSWAP evaluation.

GRID-Q demonstrates a practical quantum-HPC decision workflow for infrastructure planning, not merely a simulated QAOA benchmark.

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